TriSpectra: A Tri-Modal Architecture for Robust Audio Deepfake Detection

Abstract

The rapid spread of generative AI has led to increasingly sophisticated audio deepfakes that pose significant threats to information integrity and trust in digital media. Deepfake technology is increasingly weaponized for identity fraud and targeted harassment and misinformation. While current detection methods have strong performances on benchmark datasets, they face several limitations in real world deployment: poor generalization across different deepfake synthesis method and to real world audios, computational complexity that limits practical applications, and reliance on single modality (input data form) analysis that can reduce effectiveness for diverse types of detection cues. This project introduces a hybrid Convolutional Neural Network (CNN)-transformer architecture, TriSpectra, that uses three complementary audio data representations for deepfake detection. A residual CNN processes raw waveforms to capture temporal artifacts and an Audio Spectrogram Transformer (ASTs) analyzes Fast Fourier Transform (FFT) spectrograms and wavelet transforms to identify frequency domain inconsistencies. An attention-based fusion mechanism dynamically weighs the three modalities for the model to emphasize the most informative data representation for each audio sample. This model architecture performs significantly better with cross dataset generalization and on unseen datasets compared to current deepfake detention models.

Introduction

On the eve of a critical 2024 US primary election, twenty thousand voters across New Hampshire received a call asking them to “save their vote for November”. However, this was no ordinary phone call - it was a robocall mimicking President Joe Biden’s voice. For many, it was **indistinguishable from the real president (BBC)**.

Within the past several years, the rise of generative Artificial Intelligence (genAI) has resulted in easy to access tools that can create synthetic audiovisual media almost indistinguishable from reality (McGowan). It enables the creation of deepfakes that can be used to perpetrate fraud and scams, misinform the public, and violate bodily autonomy, all at an unprecedented scale.

Although the research community has mobilised around deepfakes, there has been a heavy skew towards visuals. In an industry-wide survey, Khanjani et al. found that out of over 150 methods and 1300 papers, **80% focus on visual deepfakes**, even though audio deepfakes are frequently used in high-stakes fraud and political manipulation (Khanjani).

Even within audio detection, most current models are designed to perform well on tightly controlled benchmark datasets but fail when faced with the unpredictability of real-world conditions. For instance, Müller et al. found a significant performance gap between the academic ASVspoof2019 evaluation data and an ‘in-the-wild’ dataset composed of audio from public domain (Müller).

This is the question that TriSpectra aims to answer: **Can we build an audio deepfake detector that performs well on real world data?**

Our approach uses **the fact that an audio clip can be represented in several different ways**. While prior work has shown that certain representations are effective in isolation (Gasenzer) **a combined architecture has not been explored**. By breaking the clip down into these representations and training Artificial Intelligence (AI) models to analyse them, we capture complementary cues that would otherwise be missed.

By combining multiple views into one system, we aim to create a model that can detect a broader range of artefacts and remain more reliable across datasets, synthesis methods, and real-world recording conditions.

Model Architecture

In plain language, what the model does is simple: it listens to the audio clip, looks at it in a few different ways, and then combines what it has learnt to determine whether the voice is real or fake.

The “three ways of looking” at the clip are:

1. The **raw sound wave** itself, which shows how the pressure changes from moment to moment.

2. A **spectrogram,** which shows what frequencies of sound are present at each time.
3. A **wavelet map**, which also shows frequencies over time, but in a way that can highlight very fast events.

By looking at the same clip in different ways, we hope to see different features that would have otherwise not been spotted.

Each view is analysed by a **specialised machine learning network that is good at spotting patterns in that form of data**. Finally, the model uses an “attention” system that compares the three views and **decides how much weight to give to each one for that particular clip**.

The key idea is that real human speech is consistent across representations, whereas deepfakes often have mismatches. **By catching these inconsistencies, the model should be able to more accurately flag synthetic voices, even if they sound human by ear.**

Formal Description of Input Representations

Spectrogram with Fast Fourier Transform (FFT)

At a high level, what we are doing is **taking the raw audio waveform and analysing which frequencies are present in it, and how they change over time.**

An audio waveform is a function of amplitude over time, but many important patterns are easier to see in terms of frequency content. Therefore, we apply a Fourier Transform to decompose the signal into a weighted sum of sinusoids:

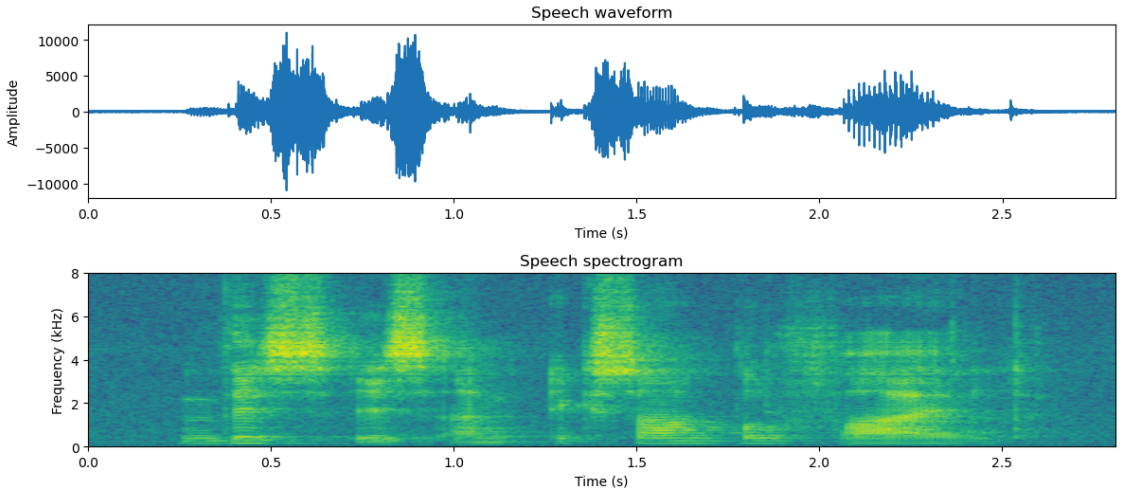
$$X(f) = \int_{-\infty}^{\infty} x(t)\, e^{-j2\pi ft}\, dt$$

where *X(f)* gives the contribution of each frequency *f*. In practice, we use the Fast Fourier Transform (FFT), an efficient algorithm for computing this decomposition on digital signals.

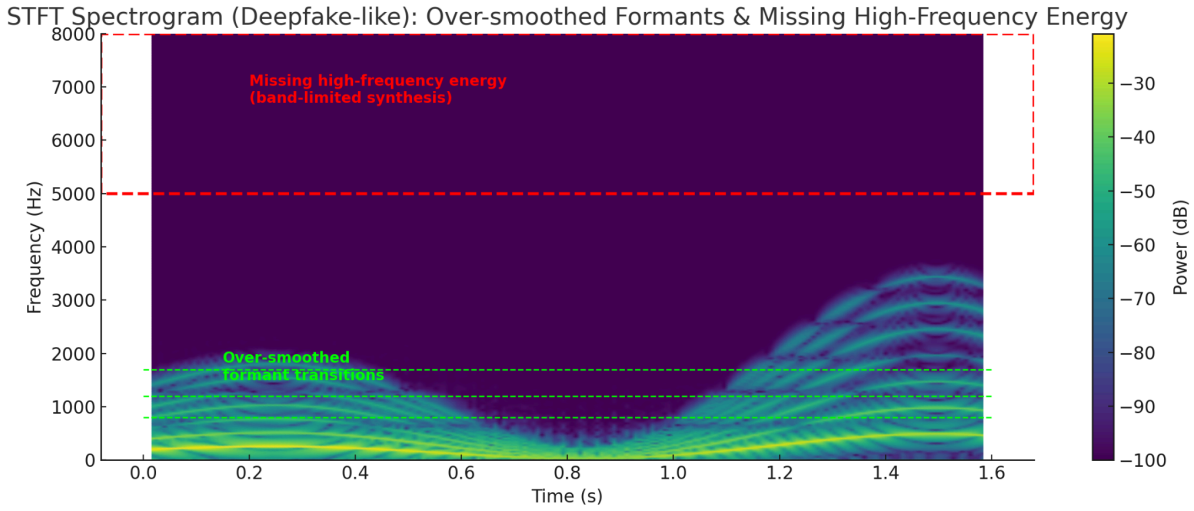
However, a single FFT of an entire speech signal would give us the global frequency content not showing us how that content evolves as we produce different vowels, consonants, and change intonation (Goyal). To handle this, we use a Short-Time Fourier Transform (STFT) in which the signal is segmented into short overlapping frames using a window function *g(t−τ)*. The transform is then applied locally:

$$\mathrm{STFT}_x(\tau, f) = \int_{-\infty}^{\infty} x(t)\, g(t-\tau)\, e^{-j2\pi ft}\, dt$$

This produces a **spectrogram**, a two-dimensional representation with time on the horizontal axis, frequency on the vertical access and intensity representing the spectral magnitude.



In this project, the **spectrogram highlights artefacts of synthesis that may not be visible in the raw waveform**, such as unnaturally smooth formant transitions or missing high frequency energy that are common in synthetic voice (Bäckström).



Wavelet Transform

Just like with the Fourier Transform, the **idea here is to look at which frequencies make up the audio signal, but this time we also want to know when particular frequencies appear and disappear.**

The Wavelet Transform decomposes audio signals into short, oscillating functions called wavelets, rather than infinite sine waves used by FFT. A wavelet starts and ends in finite time, making it suited to analysing localised features in a signal (Bäckström).

By scaling the wavelet, we obtain different frequency ranges:

- Large scales (stretched wavelets) correspond to slow oscillations, i.e. low-frequency components.
- Small scales (compressed wavelets) correspond to fast oscillations, i.e. high-frequency components.

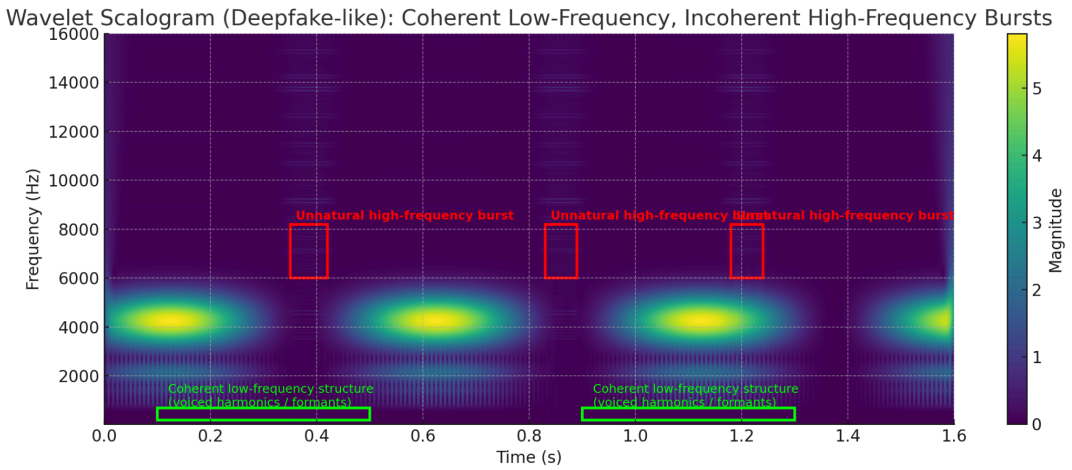
Sliding these scaled wavelets across the signal produces a time-frequency map, where time is on the x-axis and frequency (inversely related to scale) is on the y-axis, with the colour intensity showing how strongly the signal matches the wavelet at that point.

Formally, the Continuous Wavelet Transform (CWT) of a signal x(t) with mother wavelet ψ(t) is given by

$$W_x(a,b) = \int_{-\infty}^{\infty} x(t)\,\frac{1}{\sqrt{|a|}}\,\psi^*\left(\frac{t-b}{a}\right)dt$$

where a is the scale and b is the time shift. In practice, we use the Discrete Wavelet Transform (DWT) at a set of scales and positions, producing a hierarchical decomposition of low-frequency and high frequency coefficients.

For deepfake detection, this multi-resolution property is crucial. As an illustration, synthetic voices often **capture low-frequency formants correctly, but high-frequency details such as fricatives or sharp transients are harder to generate**. Wavelet maps expose these mismatches, providing complementary information to the FFT spectrogram (Goki).



Analysing Representations with Neural Networks

In order to learn which aspects of an audio clip indicate a deepfake and which indicate real speech, we train **AI Models.** These models work by being shown thousands of labelled examples and gradually adjusting millions of internal parameters to reduce mistakes. Over time, they “learn” to recognise which combinations of patterns are typical of human speech and which come from a synthetic generator (LeCun).

To analyse the **raw waveform,** we use a **Convolutional Neural Network (CNN).** A CNN applies many small filters that slide across the signal, each turned to detect certain features - for instance, tiny irregularities in timing, abrupt changes in amplitude or unnatural phase

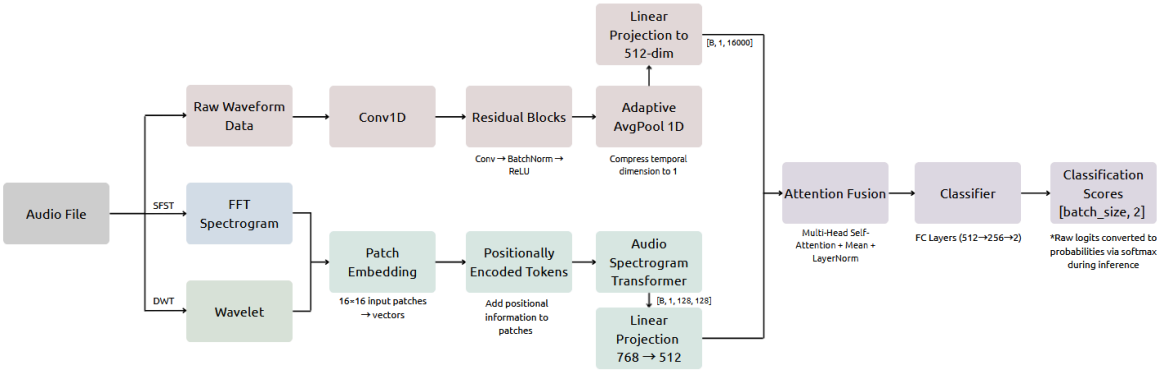
shifts. Each filter detects something different: one might learn to spot jitters, and another might pick up unnatural pauses (LeCun).

For the **wavelet maps and FFT spectrograms,** we use a Transformer, in particular an **Audio Spectrogram Transformer (AST)** (Gong)**.** Instead of filters, the transformer breaks the spectrogram into small time-frequency patches, which are represented as vectors. It then uses a process called **attention**, which is a way for the model to learn how different parts of the spectrogram engage with each other. In the context of deepfake detection, this is vital because many **synthetic voices fail to maintain these long-range relationships, leaving subtle inconsistencies that the AST can uncover.**

Once the raw waveform, spectrogram, and wavelet map have each been analysed by their own encoder, the model needs to combine these three perspectives into a single decision. A simple approach would be to just average the results or glue them together, but this often loses important information. Instead, we use a method called **attention fusion** (Gong).

This allows the model not only to decide which of the three views deserves the most weight for a given clip, but also to compare them directly. For example, if the waveform shows a phase irregularity that should correspond to a distortion in the spectrogram but does not, that mismatch itself is a useful signal of deepfake audio. By actively looking for agreement and disagreement across the three views, the model can adapt its focus and draw on whichever features are most informative. **This makes the final decision more flexible and, crucially, more generalisable to new types of deepfakes that were not seen during training.**

Summary



Overview of the TriSpectra architecture. ***Three input representations*** *(raw waveform, FFT spectrogram, and wavelet transform) are* ***encoded separately*** *and* ***fused with attention*** *before classification.*

Results

For full technical details on model implementation, visit the Github Repository. You can also find the model weights here or try a demo here.

The model was trained on three datasets:

1. ASVspoof 2019 - A standard academic benchmark consisting of real and spoofed speech generated with diverse algorithms.
2. The Fake-or-Real (FoR) dataset - Normalised and recorded training sets.
3. The In-the-Wild dataset - Containing real and fake audio from 58 celebrities and politicians on publicly available sources. **15% of the dataset was left for testing**.

Evaluation Metrics

The model is evaluated on several metrics, where TP = true positives, FP = false negatives, TN = true negatives, and FN = true positives.

Metric	**Measured Value**	**Calculation**
Accuracy	Proportion of true results (both true positives and true negatives) in the dataset.	$\frac{TP+TN}{TP+TN+FP+FN}$
Precision	Proportion of correctly identified deepfakes relative to the total number of images classified as deepfakes.	$\frac{TP}{TP+FP}$
Recall	Proportion of actual deepfakes that are correctly identified by the model.	$\frac{TP}{TP+FN}$
F1 Score	Harmonic mean of precision and recall	$2 \times \frac{Precision \bullet Recall}{Precision+Recall}$
EER	The point at which false positive rate is equal to the false negative rate	The point at which the ROC curve intersects the line FPR = 1 - TPR

Results

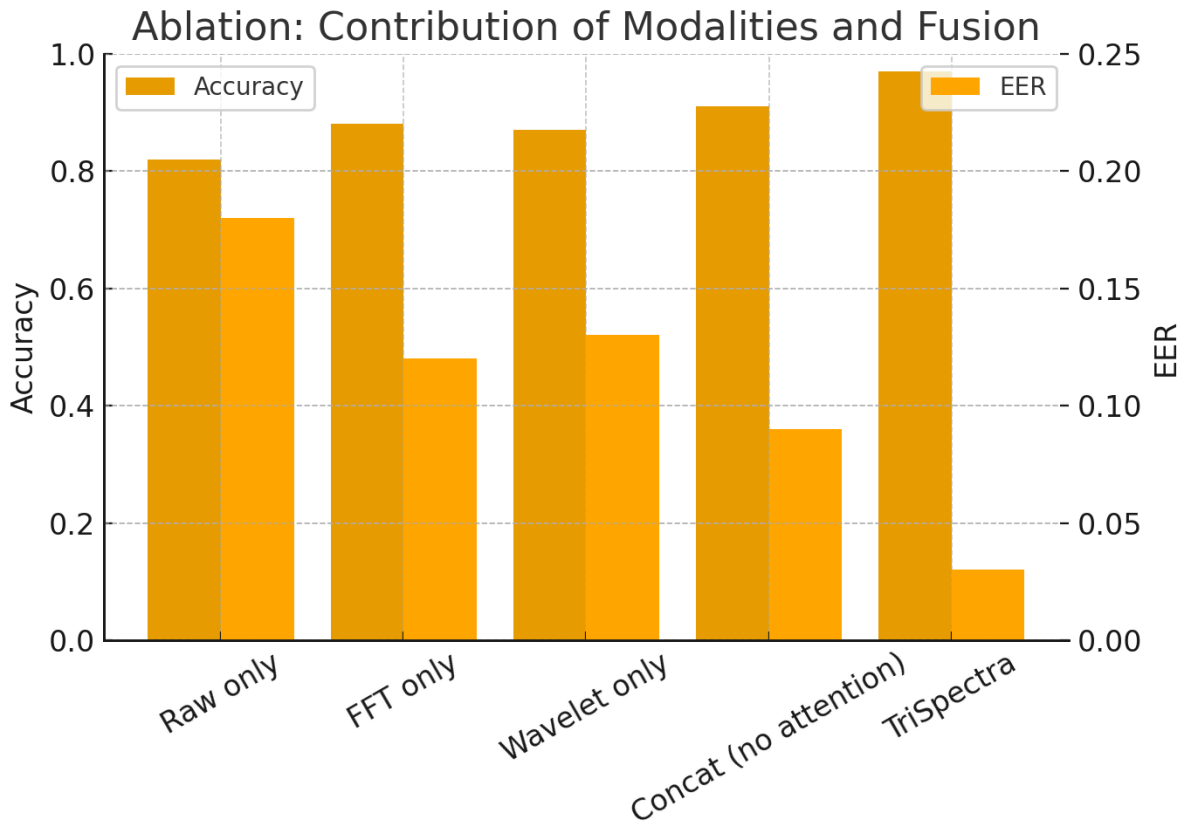
Metric	**Fake-or-Real**	**ASVspoof 2019**	**WaveFake**	**LibriSVoC**
Accuracy	0.975	0.965	0.971	0.965
F1 Score	0.975	0.960	0.965	0.910
Precision	0.970	0.955	0.976	0.900
Recall	0.980	0.965	0.955	0.920
AUC	0.980	0.970	0.966	0.935
EER	0.020	0.025	0.029	0.037

TriSpectra achieved high performance across datasets, with accuracies between 96% and 98% . The F1 scores, precision, and recall values are similarly strong, showing that the model not only catches most fake clips (high recall) but also avoids wrongly labelling too many real clips as fake (high precision). The equal error rate (EER), where false positives and false negatives balance, is also very low (2–4%), highlighting the accuracy of the model.

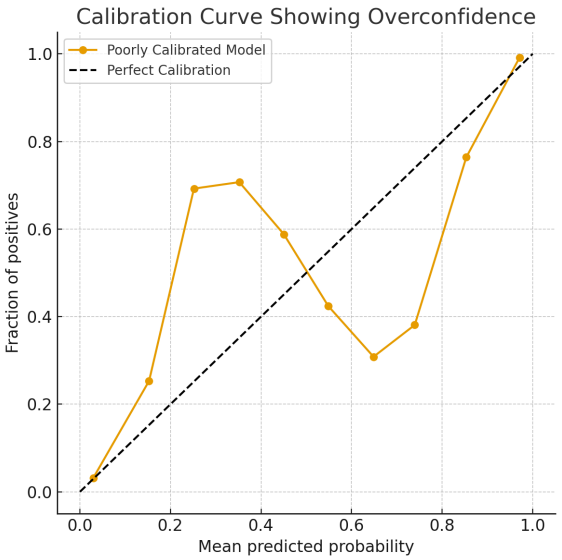
Comparative Analysis

	In-the-wild		**WaveFake**		**LibriSeVoc**	
	Accuracy	EER%	Accuracy	EER%	Accuracy	EER%
TriSpectra	**0.9402**	5.98	0.9710	2.900	**0.965**	3.70
RawNet2	0.5321	46.792	0.9416	5.839	0.5119	48.807
Wave2Vec 2	0.821	17.899	**0.9874**	1.259	0.8296	24.763
WavLM	0.829	17.08	N/A		N/A	

TriSpectra performs well on **“in-the-wild” audio**, where other models collapse. On this more difficult dataset, it reaches 94% accuracy with an error rate of just 6%. In contrast, RawNet2 - a model trained directly on waveforms - drops to almost random performance (53% accuracy). Pre-trained transformer models such as Wav2Vec2 and WavLM perform better, but still struggle: their accuracies fall to 82–83% and their error rates climb to around 18%. TriSpectra also maintains strong results on LibriSeVoc, a particularly challenging dataset, with 96% accuracy compared to RawNet2’s 51% and Wav2Vec2’s 83%.



This figure shows how well the model performs when using each input representation on its own, compared with combining them. Using only the raw waveform gives the weakest results, while FFT and wavelet spectrograms each perform better but still miss key cues. However, **TriSpectra with attention-based fusion clearly outperforms all alternatives, achieving the highest accuracy and lowest error rate.**



A calibration curve compares the model's predicted probabilities with how often it is actually correct. The dashed diagonal line shows perfect calibration: if the model predicts "70% fake," it should be right 70% of the time. Our curve falls below the diagonal, meaning the model is overconfident - when it predicts with high certainty, it is less accurate than its probabilities

suggest.

Conclusion

Audio deepfakes are only going to continue to get more realistic while also becoming easier to generate. From fraud to misinformation, we are already beginning to see the impacts across society. However, existing detection methods are falling behind: researchers predominantly focus on visual deepfakes, while current audio detectors fail to generalise to real-world data.

TriSpectra introduces a novel approach. By analysing in three different ways and training specialised neural networks on each, the model captures cues across different synthesis models. Crucially, we do not just merge these views but compare them using a final attention mechanism, allowing the model to pick up inconsistencies between timing, frequency and localised detail.

We show strong preliminary results. TriSpectra matches existing open-source models on academic datasets, but surpasses them on “in-the-wild” audio, maintaining 94% accuracy where other models suffer significantly. Ablation experiments confirm that the combination of multiple representations with attention-based fusion is key to these results. We also design an interface for users to upload their own clips and test the model.

At the same time, our evaluation shows that the model’s probability outputs are not always well calibrated. While its classifications are accurate, its confidence scores can be misleading - sometimes being overconfident when wrong. Future work includes further understanding why this is the case, as it may be a limiting factor for real world deployment.

With the importance of deepfake detection, we hope to continue to develop and stress-test TriSpectra with the aim of reaching real-world deployment. While calibration remains a limitation, TriSpectra highlights a promising path forward for the detection of audio deepfakes.

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